

MANISAI UPPALA

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PROFESSIONAL SUMMARY

Artificial Intelligence and Data Science undergraduate with strong fundamentals in Python, Deep Learning, and Machine Learning systems. Experience architecting deep learning medical computer vision pipelines (CNN ensembles), quantum-inspired quantitative analytics engines, and multi-agent LLM systems. Certified in AI Fundamentals and Linux Essentials. Seeking an AI/Data Science internship or entry-level role to apply advanced technical skills to production challenges.

EDUCATION

Velapudi Ramakrishna Siddhartha Engineering College <i>Bachelor of Technology (B.Tech) in Artificial Intelligence & Data Science</i>	Vijayawada, India 08/2024 – Present
Sri Chaitanya Junior College <i>Intermediate (U.G) – MPC</i>	Vijayawada, India 08/2022 – 04/2024
Sri Chaitanya School <i>Secondary School Certificate (SSC)</i>	Avanigadda, India 2016 – 2022

PROJECTS

- NeuroNova — Deep Learning Brain MRI Diagnostic Platform** | TensorFlow, ResNet50, MobileNetV2, Next.js 09/2026
- Architected a multi-CNN soft-voting ensemble combining fine-tuned **ResNet50, MobileNetV2, and VGG16** to classify brain MRI scans across 4 pathologies (*Glioma, Meningioma, Pituitary, No Tumor*).
 - Engineered **Hierarchical Pathology Gating** and temperature calibration ($T = 0.45$), eliminating multi-class false-negative vote splitting and elevating tumor sensitivity on difficult brainstem gliomas to **91.07%**.
 - Validated out-of-distribution (OOD) generalization against independent clinical archives, achieving **97.17% specificity** on healthy brain scans and up to **96.13% subtype diagnostic confidence**.
 - Deployed a full-stack production interface on **Vercel** with a containerized **Docker** backend, featuring real-time **Shannon Entropy Certainty Index** scoring and dynamic RESTful inference.
- Q-FinOpt — Quantum-Inspired Mutual Fund Portfolio Optimizer** | Python, QAOA, GBM, FastAPI, Kotlin 06/2024
- Built an end-to-end quantitative pipeline processing 419,188 daily AMFI records across 296+ mutual funds, engineering 12 risk-adjusted momentum, volatility, and CAPM alpha factors.
 - Trained a Gradient Boosting return forecaster achieving **74.01% directional accuracy** ($p < 0.001$) and **+0.89%/month active alpha** over benchmark indices.
 - Mapped fund selection to a QUBO Ising Hamiltonian and executed a 2-layer QAOA circuit ($O(p \cdot N^2)$), achieving a **2.22 Sharpe ratio** and **+61.40% 36-month backtest return** (vs. NIFTY 50 +42.10%).
 - Architected a FastAPI backend and native Android app (Jetpack Compose); pilot-validated across 45 client portfolios with a **9.6/10 client satisfaction score**.
- Multi-Agent Research Assistant** | LangGraph, Groq, Tavily API, Streamlit, Python 04/2024
- Built a 4-agent LangGraph workflow (*Planner, Researcher, Critic, Writer*) with feedback loops capped at 3 iterations to autonomously generate verified, source-attributed technical research reports.
 - Integrated Groq LLaMA models and Tavily search to generate structured Markdown dossiers with citation linking, robust retry handling, and rate-limit mitigation.
 - Deployed on Streamlit Community Cloud featuring Quick, Standard, and Deep research modes with real-time agent-state streaming and automated report downloads.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, HTML5, CSS3
Libraries & Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Pillow, FastAPI, Flask, Next.js 14, React
Machine Learning & AI: Computer Vision (CNNs, Transfer Learning, Ensemble Learning), Gradient Boosting, XGBoost, Time Series Forecasting, Monte Carlo Simulation (GBM), Quantum QAOA, LangGraph Multi-Agent Systems, Prompt Engineering
Tools, Platforms & DevOps: Docker, Git, GitHub, Linux, Vercel, Render, Streamlit, Android Studio, Google Colab, Jupyter, Vertex AI, MySQL

CERTIFICATIONS

- Prompt Design in Vertex AI** – Google Cloud (Effective multimodal prompt engineering)

- **Artificial Intelligence Fundamentals** – IBM SkillsBuild (Machine Learning, Computer Vision, NLP & Deep Learning)
- **Inspect Rich Documents with Gemini Multimodality & Multimodal RAG** – Google Cloud
- **Linux Essentials** – Cisco / Credly (Core OS architecture, bash scripting & system management)